Week 10

TECHNOLOGY ARCHITECTURE DESIGN PRINCIPLE

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5 PRINCIPLES IN EA ARCHITECTURE

- **Understandable**, the intention of the principle is clear so that violations are minimised.
- **Robust** - enable good quality decisions to be made, and enforceable policies and standards to be created.
- **Complete** - every potentially important principle governing the management of information and technology for the organisation is defined.
- **Consistent** - principles should not be contradictory to the point where adhering to one principle would violate the spirit of another.
- **Stable** - principles should be enduring, yet able to accommodate changes.

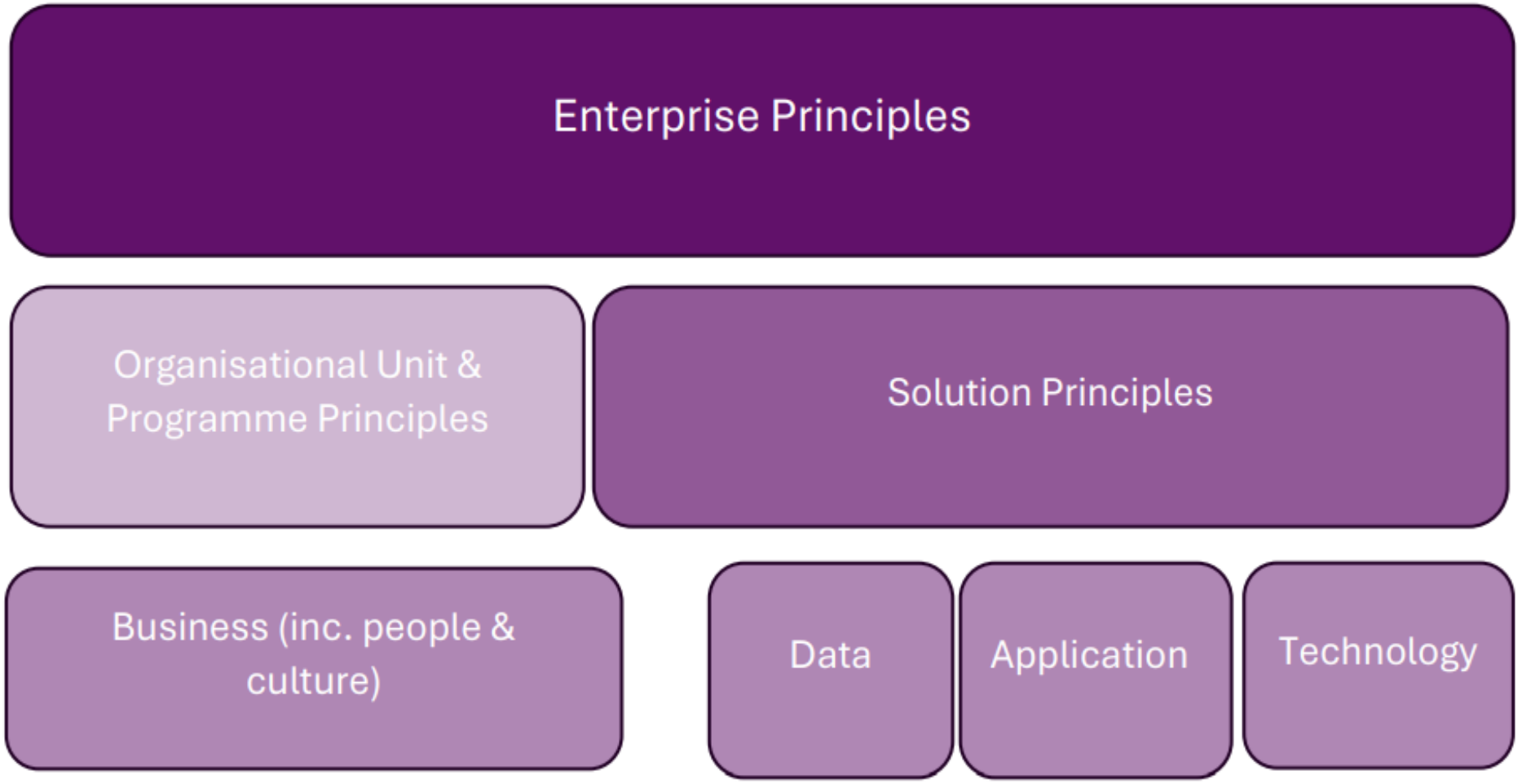


Diagram 1. Hierarcy of architecture design principles



1. Enterprise Principles (Top Level)

- Represent the **highest-level guidance** governing the entire organization.
- Provide overarching rules and values that shape all architectural decisions.
- Ensure alignment with the organization’s mission, vision, and strategic objectives.
- Influence every subsequent layer of principles in the hierarchy.



3. Solution Principles

- Provide the rules and guidelines for **designing and implementing solutions** (systems, services, platforms).
- Ensure that solutions adhere to enterprise-wide standards and programs-specific requirements.
- Guide decisions about system architecture, integration, user experience, and technology adoption.
- Connect the high-level organisational intentions to concrete design outcomes.

2. Organisational Unit & Programmed Principles

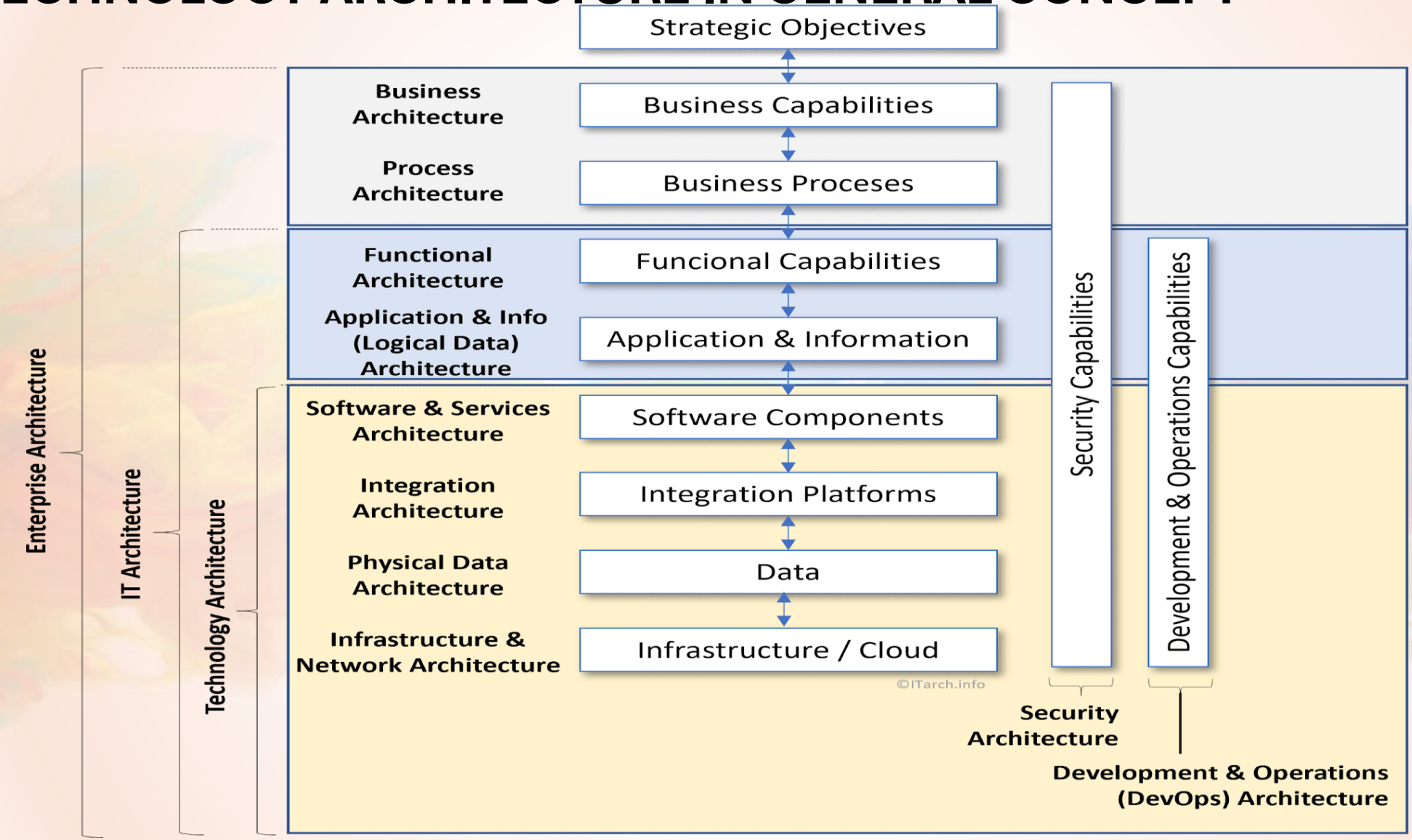
- Derived from the enterprise principles but adapted for **specific departments, units, or programs**.
- Ensure that each organizational unit operates within the broader enterprise framework.
- Provide direction tailored to the unique needs of projects, programs, or divisions.
- Maintain consistency across multiple business units while allowing contextual flexibility.



4. Domain Principles (Bottom Layer)

- These principles operate at a **more detailed, domain-specific level**, including:
  - **a. Business Principles (including people & culture)**
    - Govern how business processes, people, culture, and organizational behavior should align with the architecture.
    - Provide direction for business transformation, operational models, and stakeholder engagement.
  - **b. Data Principles**
    - Define how data should be collected, managed, shared, secured, and used.
    - Support interoperability, data quality, governance, and lifecycle management.
  - **c. Application Principles**
    - Guide how applications are designed, integrated, deployed, and maintained.
    - Support modularity, scalability, reusability, and interoperability.
  - **d. Technology Principles**
    - Framework for selecting, managing, and standardizing technology platforms, tools, and infrastructure.
    - Ensure robustness, security, efficiency, and future scalability.

TECHNOLOGY ARCHITECTURE IN GENERAL CONCEPT

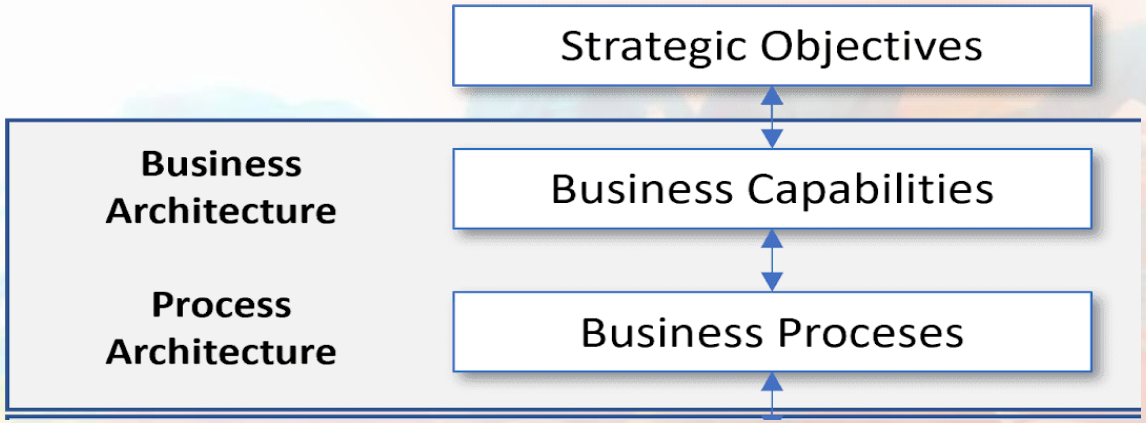


Strategic Objectives (Top Layer)

- The highest-level direction of the organization.
- Defines *why* the enterprise exists and *what major goals it must achieve*.
- All other architectural layers must align with these objectives.

Examples:

- “Increase digital service adoption by 40%.”
- “Reduce operational cost by automating processes.”
- “Strengthen cybersecurity maturity across all departments.



Business Architecture & Process Architecture

This layer defines:

- **Business Capabilities:** What the business must be able to do.
- **Business Processes:** How the organization performs these capabilities.

Purpose

- Ensures technology solutions support actual business needs.
- Establishes a clear mapping from strategy → business operations.

Examples of Business Capabilities

- Customer Onboarding, Order Management, Billing & Payments, Supply Chain Operations.
- **Examples of Business Processes:** Verify customer identity, Approve loan application, Handle product returns

Functional Architecture & Application / Information (Logical Data) Architecture

This layer translates business capabilities into:

- **Functional Capabilities:** System-level functions required to enable business processes.
- **Application & Information Architecture:** Logical structure of applications and the data they handle.

Purpose

- Ensures systems are logically organized.
- Defines how information flows across the enterprise.

Examples of Functional Capabilities: Authentication, Payment authorization, Customer profile management.

Examples of Applications / Logical Data: CRM (Customer Master Data), KYC Verification System, Logical data entities such as *Customer*, *Account*, *Transaction*, *Document*

Technology Architecture Layer

- **A. Software & Services Architecture**
  - Actual software components that implement the functional capabilities.
  - **Examples:** auth-service microservice, Notification service, Web portal frontend, Back-end API service
- **B. Integration Architecture**
  - Defines how applications and components communicate.
  - **Examples:** API Gateway, Message broker (Kafka, RabbitMQ), Enterprise Service Bus (ESB), Webhooks,
- **C. Physical Data Architecture**
  - How data is stored physically.
  - **Examples:** PostgreSQL, MySQL (transaction data), Snowflake / BigQuery (analytical data), Amazon S3 / Azure Blob Storage (document storage), Replication, backup schedules
- **D. Infrastructure & Network Architecture**
  - The platform where all systems run: compute, storage, network, cloud, servers.
  - **Examples:** AWS / Azure / Google Cloud, Kubernetes clusters (EKS, AKS, GKE), Load balancers, VPC networks, CDN for content delivery

5. Cross-Layer Architecture: Security Capabilities

- Security applies to every layer: business, application, data, and infrastructure.
- **Key Components:** Identity & Access Management (IAM), Encryption, PKI, Zero Trust architecture, Threat detection systems
- **Examples:** MFA for all users, TLS encryption for all services, Role-Based Access Control (RBAC), WAF (Web Application Firewall)

6. Cross-Layer Architecture: Development & Operations (DevOps) Capabilities

- DevOps governs how software is built, deployed, monitored, and improved continuously.
- **Examples:** CI/CD pipelines (Jenkins, GitHub Actions), Containerization & orchestration, Monitoring dashboards (Prometheus + Grafana), Automated rollback and deployment strategies

Software Components

- auth-service
- kyc-microservice
- crm-api
- React-based web/mobile UI

Integration Architecture

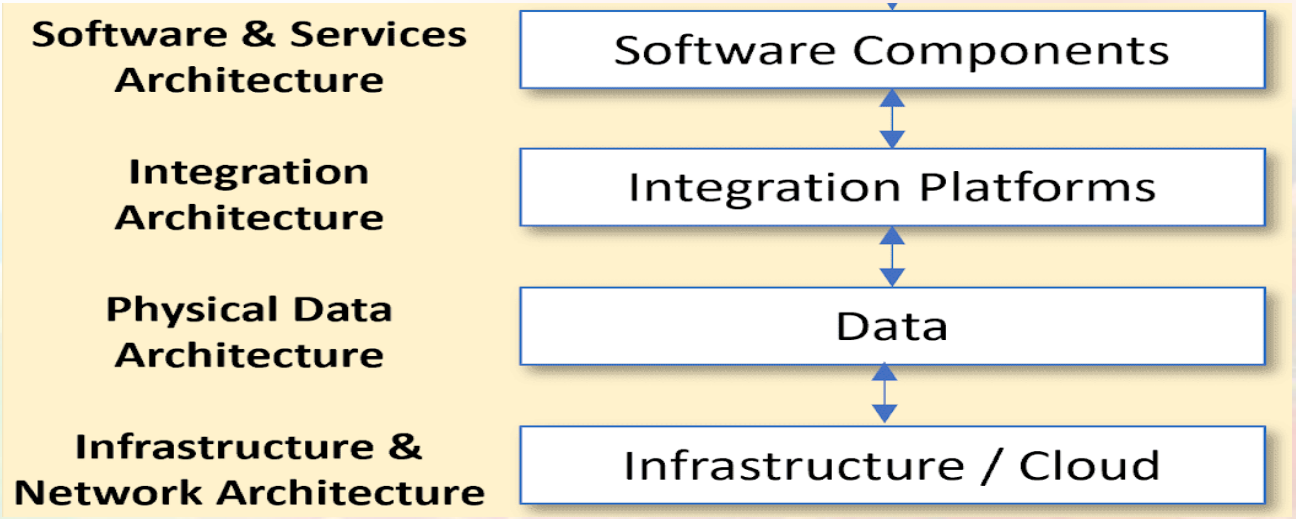
- API Gateway routes /signup
- Kafka topic for asynchronous verification
- Webhooks for success notification

Physical Data

- PostgreSQL stores customer accounts
- S3 stores scanned documents
- Data warehouse stores analytics

Infrastructure Layer

- Runs on AWS: EKS cluster, Load Balancer, VPC
- Auto-scaling to handle spike requests



TECHNOLOGY DIAGRAM

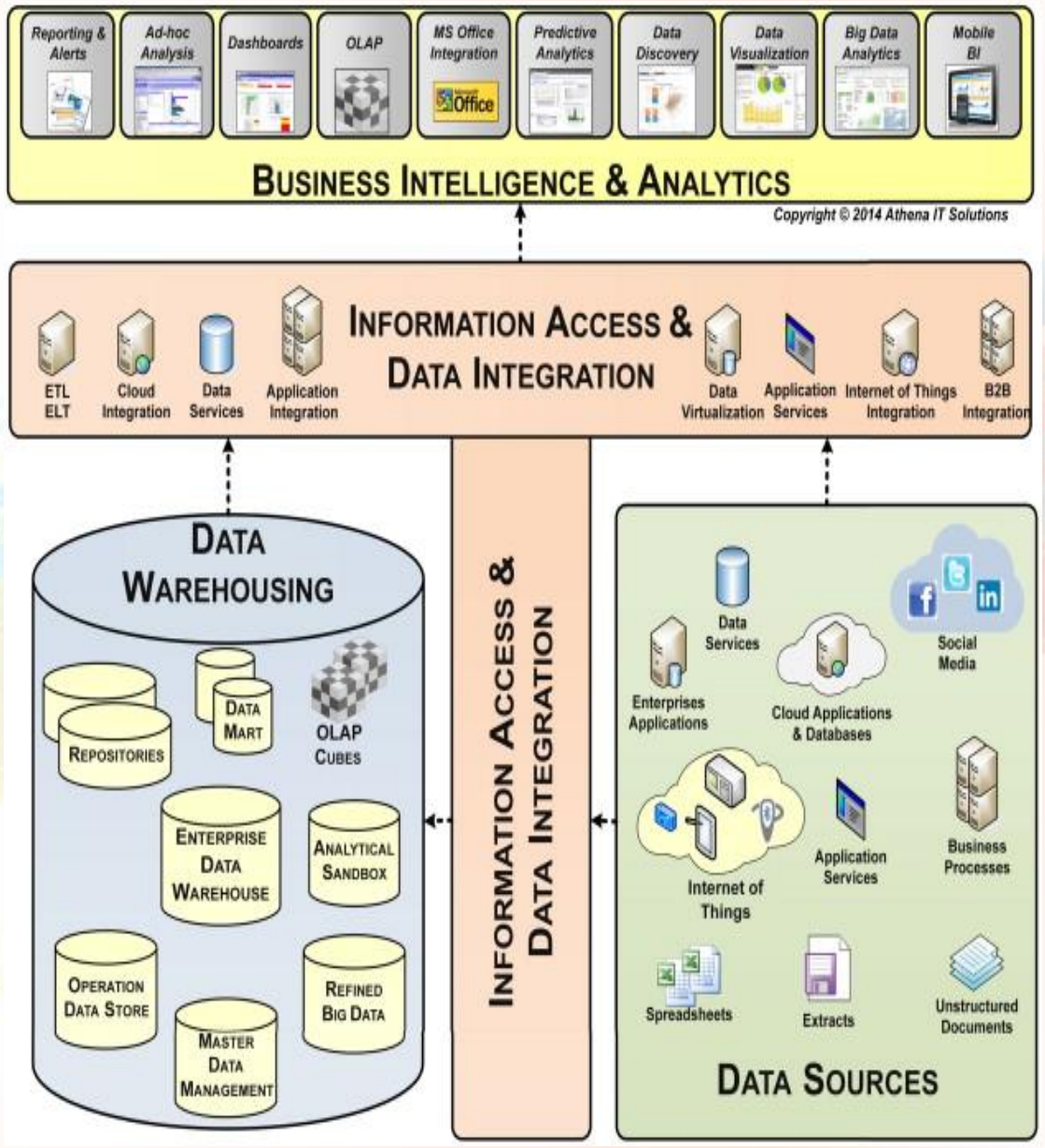
Data Sources
•Data originates from various systems, IoT devices, files, social networks, and databases.

Integration Layer
•Data is connected, standardized, transformed, or virtualized for organizational use.

Data Warehouse
Data is stored, cleansed, structured, and made analytics-ready.

Access & Integration Layer
Data is fed into BI tools through ETL, services, or APIs.

Business Intelligence & Analytics → Data is interpreted, visualized, and used for reporting, prediction, and decision-making.



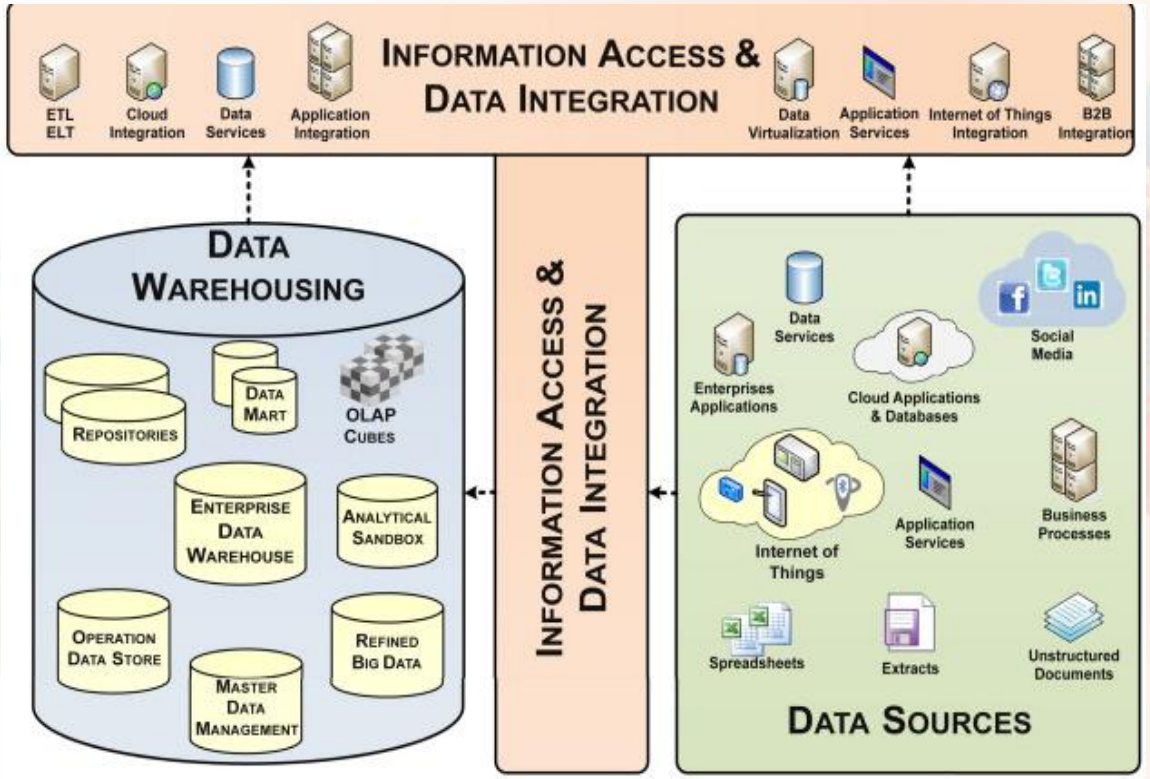
Business Intelligence & Analytics



- The top layer provides analytical capabilities that turn raw data into actionable insights for business users and executives.
- It includes a wide range of analytical tools such as:
  - **Reporting & Alerts:** Automated reports and event-driven notifications.
  - **Ad-hoc Analysis:** User-driven exploration of data using flexible tools.
  - **Dashboards:** Visual KPIs and operational metrics updated in real time.
  - **OLAP Analysis:** Multidimensional slicing and dicing of data (time, region, product).
  - **Predictive Analytics:** Using statistical models and machine learning for forecasting.
  - **Data Discovery:** Identifying patterns, anomalies, and correlations.
  - **Data Visualization:** Presenting insights using charts, graphs, and interactive visuals.
  - **Big Data Analytics:** Advanced analysis on large-scale datasets.
  - **Mobile BI:** Accessing dashboards and analytics via phones and tablets.
- These tools help organizations make better decisions, optimize operations, and identify new opportunities.

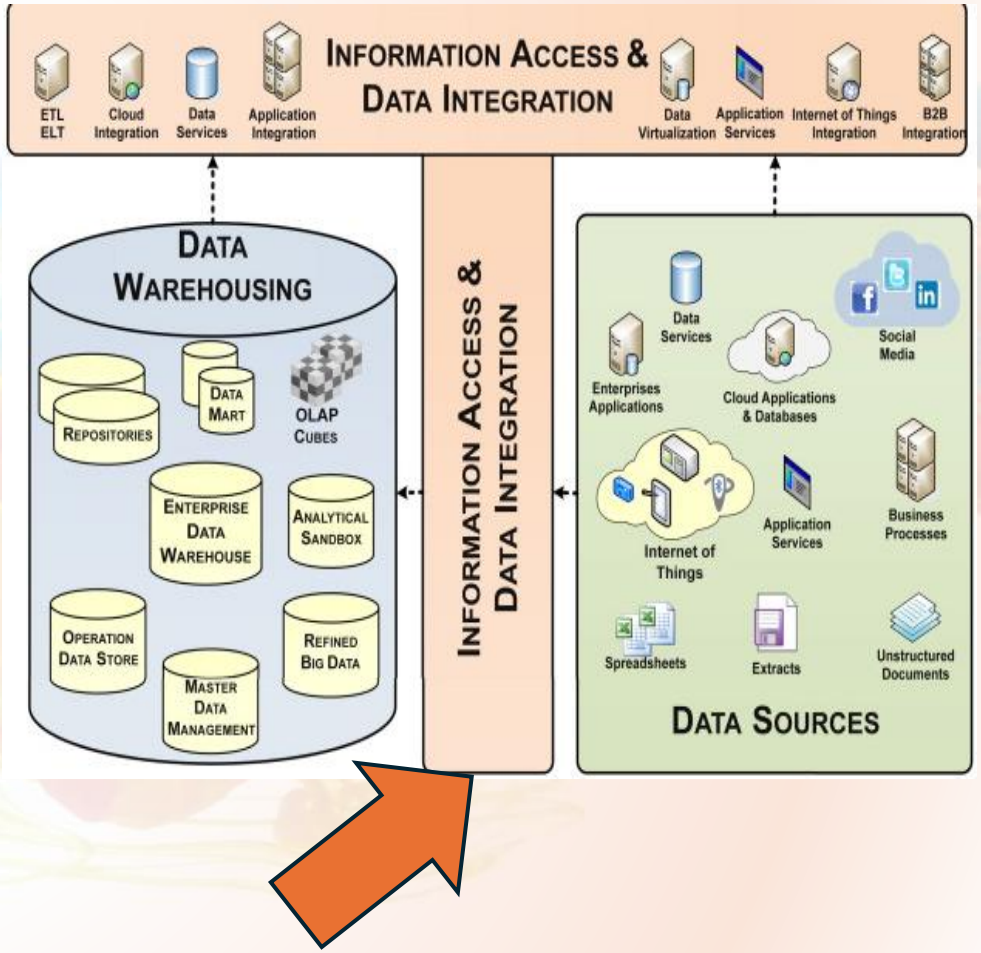
Information Access & Data Integration

- This layer is responsible for **connecting, harmonizing, and standardizing** data coming from various sources so that it can be used consistently across the organization.
- It includes technologies such as **data services, application services, data virtualization, cloud integration, IoT integration, and B2B system integration**.
- The purpose of this layer is to ensure that data can be accessed securely, reliably, and in real time, regardless of where the data originally resides.
- It acts as the bridge between raw data sources and advanced analytics systems.



Information Access & Data Integration

- After the data warehouse processes and organizes data, this upper integration layer prepares the data for **consumption by business intelligence tools**.
- It leverages technologies such as:
  - **ETL/ELT pipelines** to move and transform data.
  - **Cloud integration tools** to handle hybrid or multi-cloud environments.
  - **Data services and APIs** to provide access to data in real time.
  - **Application integration frameworks** to connect operational systems with analytical platforms.
- This layer ensures that high-quality data flows smoothly into BI systems and reporting environments.



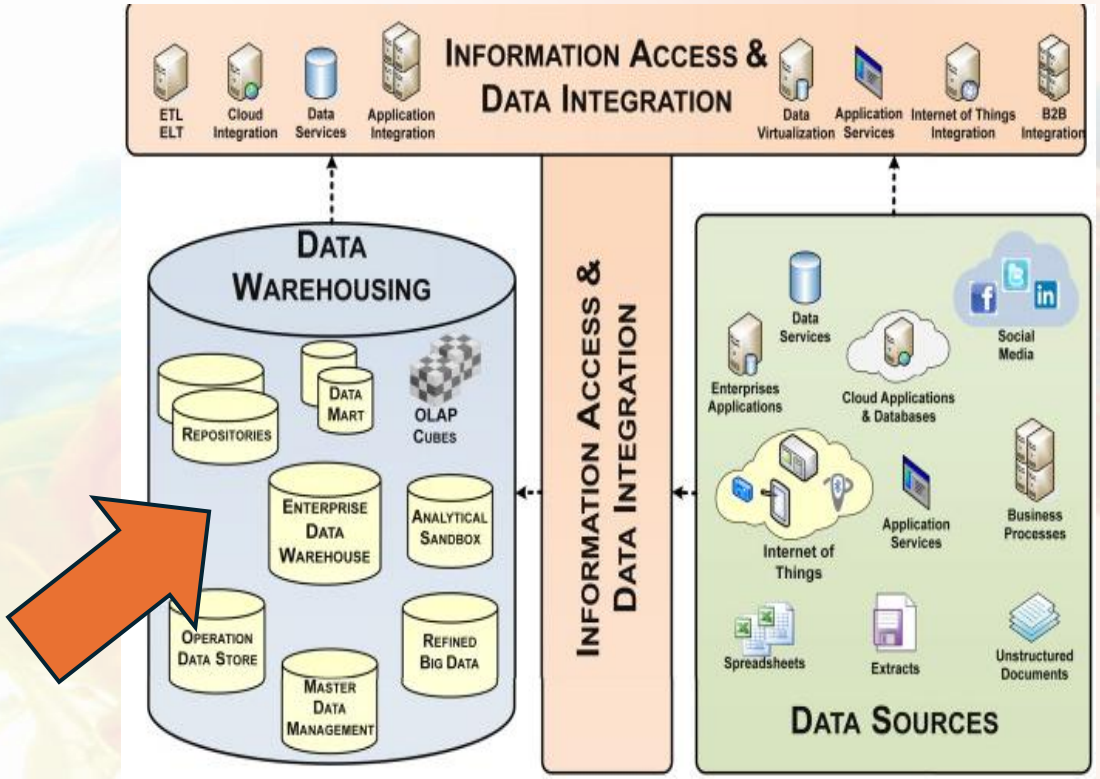
Data Warehousing

Data Warehousing is the centralized storage layer where integrated data from the entire organization is kept for reporting, analytics, and decision-making.

It includes several internal components:

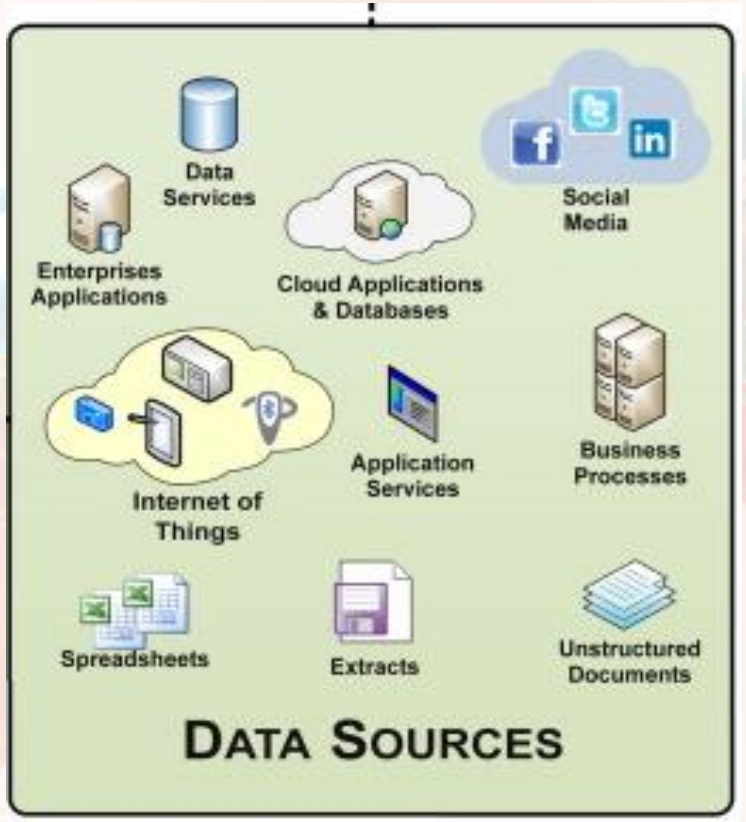
- **Operational Data Store (ODS):** Stores near-real-time operational data.
- **Master Data Management (MDM):** Defines and manages core entities such as customers, products, and vendors.
- **Enterprise Data Warehouse (EDW):** The central repository for historical, structured, and integrated enterprise-wide data.
- **Data Marts:** Subject-specific storage for departments such as marketing, finance, or HR.
- **OLAP Cubes:** Enable multidimensional analysis across time, geography, product line, etc.
- **Analytical Sandbox:** A working area for data scientists to explore and experiment.
- **Refined Big Data:** Cleaned and processed big data ready for advanced analytics.

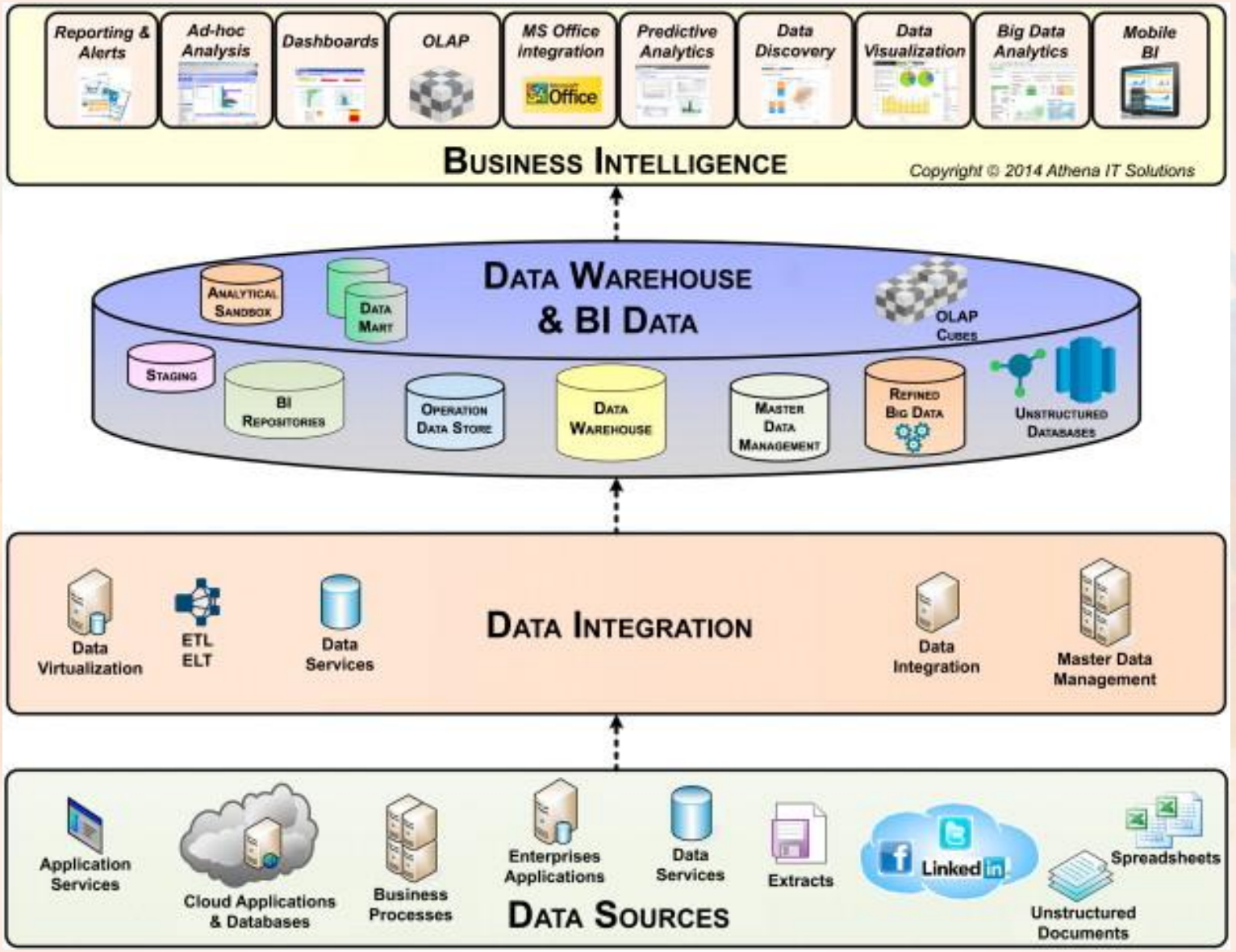
The purpose of the data warehouse is to ensure clean, consistent, historical data is available for strategic and tactical analysis.



Data Sources

- **Data Sources** represent all originating points where data is created or collected. These sources may include *enterprise applications*, *cloud applications and databases*, *Internet of Things (IoT) devices*, *business process systems*, *external data extracts*, *spreadsheets*, *social media platforms*, and *unstructured documents* (emails, PDFs, logs, images).
- These sources differ in format (structured, semi-structured, unstructured), frequency (batch or real-time), and level of complexity.
- Organizations must integrate these diverse data sources to gain a complete and accurate understanding of business activities.




Reporting & Alerts
Ad-hoc Analysis
Dashboards
OLAP
MS Office Integration
Predictive Analytics
Data Discovery
Data Visualization
Big Data Analytics
Mobile BI
BUSINESS INTELLIGENCE
Copyright © 2014 Athena IT Solutions
DATA WAREHOUSE & BI DATA
ANALYTICAL SANDBOX
DATA MART
OLAP CUBES
STAGING
BI REPOSITORIES
OPERATION DATA STORE
DATA WAREHOUSE
MASTER DATA MANAGEMENT
REFINED BIG DATA
UNSTRUCTURED DATABASES
Data Virtualization
ETL ELT
Data Services
DATA INTEGRATION
Data Integration
Master Data Management
Application Services
Cloud Applications & Databases
Business Processes
Enterprises Applications
Data Services
Extracts
Linked in
Spreadsheets
Unstructured Documents
DATA SOURCES

Data Integration Layer

- Converts fragmented raw data into *clean, consistent, structured* data.
- Ensures data quality and reliability.
- Prepares data for:
  - dashboards
  - reporting
  - advanced analytics
  - machine learning
- Serves as the critical foundation for Business Intelligence.

Data Warehouse / BI Data Components

- These components store, organize, and prepare data for different types of analytical workloads.

Key Components

- **Data Warehouse:** Central repository for historical, integrated, and structured enterprise data.
- **Data Mart:** Subset of the Data Warehouse optimized for a specific department (Finance, Marketing, Sales).
- **Operational Data Store (ODS):** Stores near-real-time operational data for fast reporting.
- **Analytical Sandbox:** Dedicated area for data scientists to experiment and model data.
- **BI Repositories:** Storage for metadata and reporting models.
- **Refined Big Data:** Cleansed and structured big data ready for analytics.
- **Unstructured Databases:** Systems storing non-structured data ready for analysis.
- **OLAP Cubes:** Multidimensional structures enabling advanced analytical queries.

Data Integration Layer

- This layer is responsible for **collecting, transforming, cleansing, and harmonizing** data from the Data Sources layer to make it suitable for analytical use.
- **Key Components**
- **Data Virtualization:** Accessing data across systems without physically moving it.
- **ETL/ELT Processes:** Extract, Transform, Load (or Load then Transform) pipelines that standardize and move data.
- **Data Services:** APIs that provide unified access to integrated or processed data.
- **Data Integration:** Combining data from multiple systems into a consistent, usable format.
- **Master Data Management (MDM):** Ensuring consistent definitions and values for key data entities (e.g., Customer, Product).

Data Sources Layer

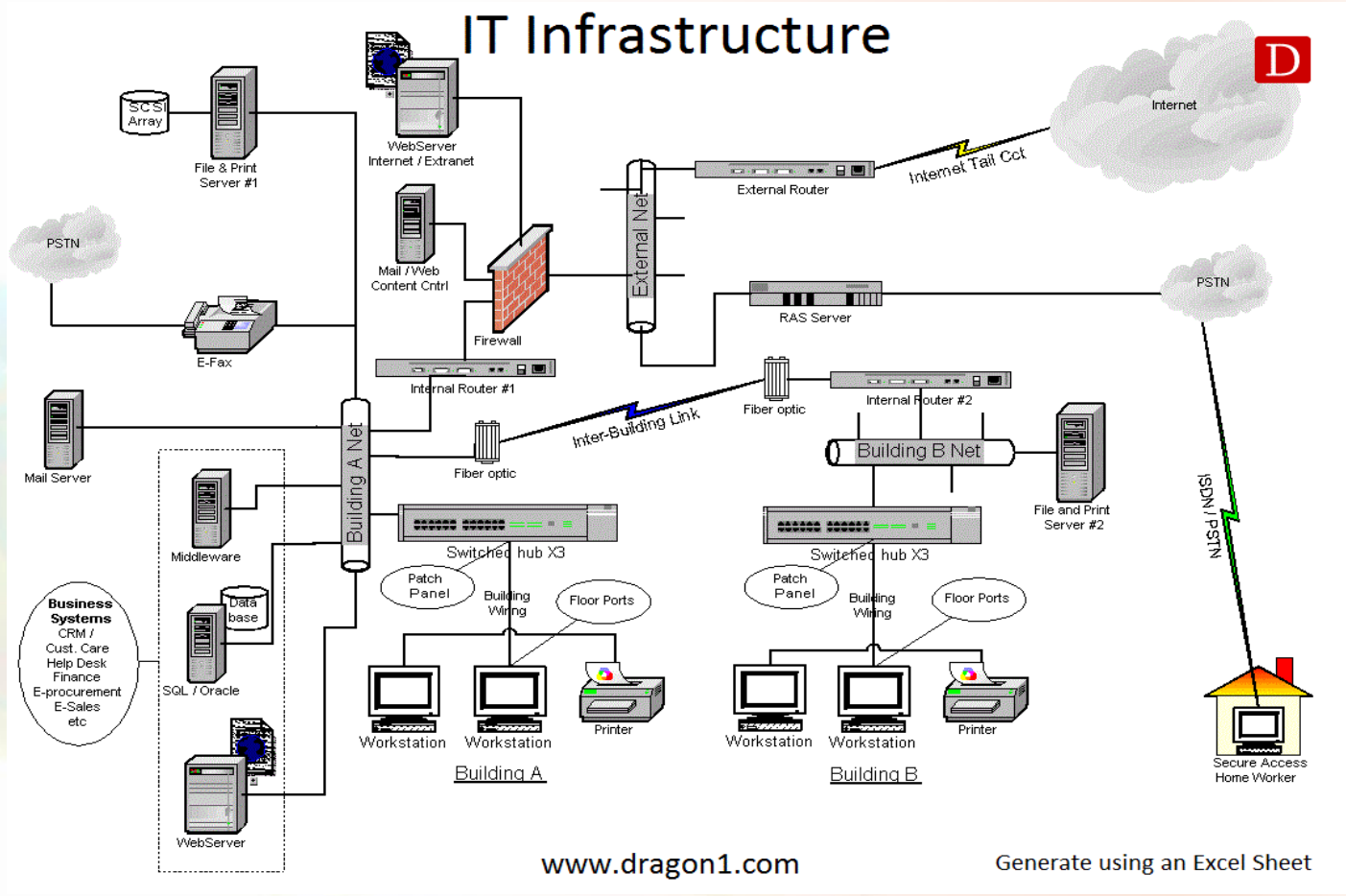
- This is the foundational layer where all organizational data originates. It includes a wide variety of internal and external systems that generate, store, or process data.

Key Components in the Layer

- **Enterprise Applications:** Core internal systems such as ERP, CRM, HRM, and financial systems.
- **Cloud Applications & Databases:** SaaS platforms and cloud-native databases used for operations (e.g., Salesforce, Google Workspace).
- **Business Processes:** Activities and workflows that generate operational data.
- **Application Services:** APIs or microservices that expose operational data in real time.
- **Data Services:** Internal services that manage access to enterprise data.
- **Extracts:** External or third-party datasets imported into the organization.
- **Social Media:** Sources like Facebook, Twitter, LinkedIn for sentiment analysis or marketing data.
- **Spreadsheets:** Data created manually by users, often used for analysis and reporting.
- **Unstructured Documents:** PDFs, emails, images, logs, notes, audio, and other non-tabular content.
- **Overall Meaning**

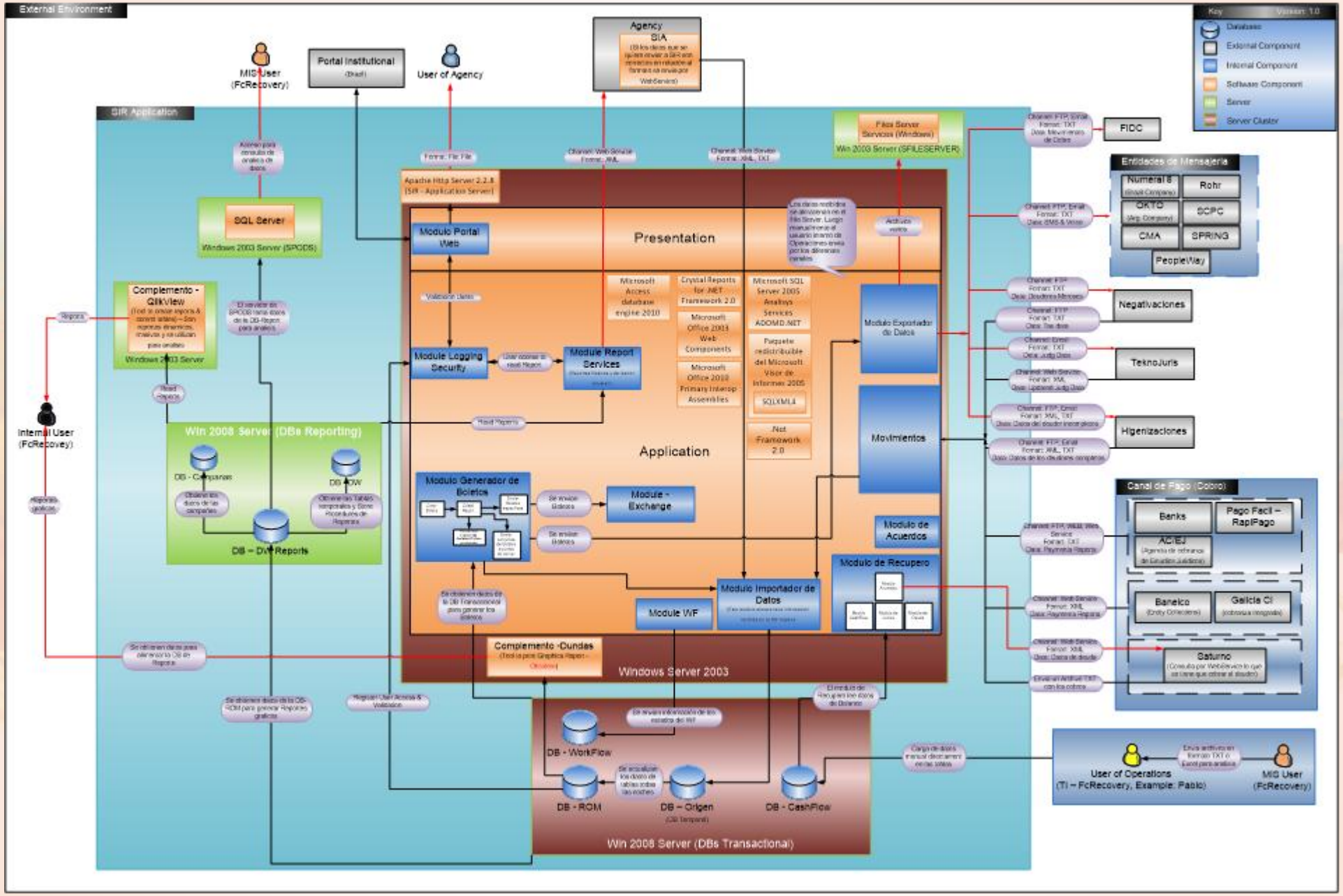
Example

- Secure network segmentation
- Dedicated server roles
- Cross-building connectivity
- Remote access capabilities.
- It provides a stable foundation for operational efficiency, data consistency, and secure communication across all departments.



Common adopters include:

- **Large manufacturing companies**
  - For ERP, production systems, supply chain management.
- **Banks and financial institutions**
  - For core banking, transaction systems, and 24/7 operations.
- **Government agencies**
  - With multiple offices and heavy internal application usage.
- **Large universities**
  - Needing academic servers, finance systems, and student portals.
- **Telecommunications companies**
  - For billing systems, CRM, and network monitoring.


External Environment
Agency
SIA
MIS User
(FcRecovery)
Portal Institutional
User of Agency
Key
Version: 1.0
Database
External Component
Internal Component
Software Component
Server
Server Cluster
SIR Application
Files Server
Services (Windows)
Win 2003 Server (SFILESERVER)
FIDC
Entidades de Mensajería
Numeral 8
Rohr
OKTO
SCPC
CMA
SPRING
PeopleWay
SQL Server
Windows 2003 Server (SPDDB)
Apache Http Server 2.2.8
(SIR - Application Server)
Presentation
Module Portal
Web
Complemento -
QlikView
Windows 2003 Server
Module Logging
Security
Module Report
Services
Microsoft
Access
database
engine 2010
Crystal Reports
for .NET
Framework 2.0
Microsoft
Office 2003
Web
Components
Microsoft
Office 2010
Primary Interop
Assemblies
Microsoft SQL
Server 2005
Analisys
Services
ADOMD.NET
Paquete
redistribuible
del Microsoft
Visor de
Informes 2005
SQLXML4
Net
Framework
2.0
Application
Module Exporter
de Datos
Movimientos
Negativaciones
TeknoJuris
Higienizaciones
Internal User
(FcRecovey)
Win 2008 Server (DBs Reporting)
DB - Campanas
DB - DW
DB – DW Reports
Modulo Generador de
Boletos
Module -
Exchange
Modulo de
Acuerdos
Modulo de Recupero
Modulo Importador de
Datos
Module WF
Canal de Pago (Cobro)
Banks
Pago Facil –
RapiPago
ACTEJ
Banelco
Galicia CI
Saturno
Complemento -Dundas
Windows Server 2003
DB - WorkFlow
DB - ROM
DB – Origen
DB - CashFlow
Win 2008 Server (DBs Transactional)
User of Operations
(TI – FcRecovery, Example: Pablo)
MIS User
(FcRecovery)

The overall purpose is to support a highly integrated enterprise platform that **handles**: large-scale data processing, reporting, workflow management, payment reconciliation, and external agency compliance. This architecture ensures data consistency, traceability, interoperability with third-party systems, and efficiency in internal operations.

Key Points

- Centralises data processing for large institutions.
- Ensures regulatory compliance through agency integrations.
- Provides real-time and batch data processing capabilities.
- Improves transparency through detailed logging and reporting.
- Supports high-volume financial or transactional operations.

Types of Organizations Using This Model

- This type of architecture is commonly found in institutions that require heavy data processing, compliance communication, and multi-department coordination. It is typical in government agencies, banks, financial platforms, insurance companies, and large-scale enterprises.

- **Key Points**
- Government administrations
- Banking and financial institutions
- Insurance companies
- Payment processing companies
- Enterprise organizations with workflow and reporting needs